



Making Shiny Dashboards Accessible for Keyboard and Screen Reader

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Land Acknowledgement

Every community owes its existence and vitality to generations from around the world who contributed their hopes, dreams, and energy to making the history that led to this moment. Some were brought here against their will, some were drawn to leave their distant homes in hope of a better life, and some have lived on this land for more generations than can be counted. Truth and acknowledgment are critical to building mutual respect and connection across all barriers of heritage and difference.

We begin this effort to acknowledge what has been buried by honoring the truth. We are standing on the ancestral lands of the Dakota people. We want to acknowledge the Dakota, the Ojibwe, the Ho Chunk, and the other nations of people who also called this place home. We pay respects to their elders past and present. Please take a moment to consider the treaties made by the Tribal nations that entitle non-Native people to live and work on traditional Native lands. Consider the many legacies of violence, displacement, migration, and settlement that bring us together here today. Please join us in uncovering such truths at any and all public events.*

*This is the acknowledgment given in the USDAC Honor Native Land Guide – edited to reflect this space by Shannon Geshick, MTAG, Executive Director Minnesota Indian Affairs Council

Introduction

Why Accessibility Matters

- 1 in 4 U.S. adults live with a disability — many use keyboards or screen readers daily.
- US Government dashboards must meet WCAG 2.1 AA standards.
- Accessible design improves usability for everyone: keyboard power users, mobile users, and more.
- Legal and compliance risk: inaccessible public-sector tools can trigger ADA complaints.
- The fixes are mostly small, big impact for modest effort.

'Fun with Irises' dashboard structure and issues

App Structure

- Three tabs — Introduction, Barplot, Histogram
- A sidebar with 4 filters (dropdowns + sliders)
- Charts generated from R's built-in iris dataset

Known Issues (pre-fix)

- No page language defined
- No chart alt text
- Slider filters lose keyboard focus
- Filter labels skipped by screen readers
- No browser tab title
- Filters show on all tabs (even irrelevant ones)

Three Categories of Issues

Screen Reader Access

- Language not defined
- No alt text on charts
- Filter labels skipped
- Charts only readable visually

Keyboard-Only Access

- Slider focus not visible
- Double-slider can't be used via keyboard

Clarity & Understanding

- No browser tab title
- Only one navigation method
- Vague filter labels
- Filters visible on wrong tabs

Tools Used to Find Issues

shinya11y

R package

Check labels, headings, and links or use the Screen Reader Wand to inspect alt text.

ANDI

Bookmarklet

Check page structure, focus order, alt text, and element labels.

NVDA

Screen Reader

Hear how your dashboard is read aloud.

WAVE

Browser tool

Visual overlay showing errors, alerts, and structural elements.

Resources used or referenced

- [shinya11y](#) R package, Ewen Henderson et al (@ewenme on GitHub)
- [Accessible Name and Description Inspector \(ANDI\)](#) bookmarklet, US Social Security Administration
- [Web Accessibility Evaluation Tool \(WAVE\)](#) browser extension, Web Accessibility in Mind (WebAIM), Utah State University
- [NonVisual Desktop Access \(NVDA\)](#) screen reader, NV Access
- [Language Subtag Lookup](#) webpage, Internet Corporation for Assigned Names and Numbers (ICANN)

'Fun with Irises' dashboard resources

[R/Medicine 2026 presentation files](#)

- Non-accessible app — Practice the fixes yourself
- Accessible app — Reference implementation with all fixes applied
- Demonstration outline — One-page summary of all topics covered
- Demonstration script and folder readme — Script describing all fixes covered

Conclusion

What we fixed

- Defined:
 - page language
 - browser tab title
- Replaced:
 - single slider to dropdown
 - double slider to text inputs
- Fixed dropdown labels

- Added:
 - table-of-contents navigation links
 - navigation instructions
 - dynamic chart descriptions
 - conditional filters per tab
 - data tables for charts
 - reactive alt text to charts

Build It In From the Start: Template tricks

- Include `tags$html(lang = 'en')` and `tags$title('...')`.
- Use `selectInput(selectize = FALSE)` for all filter dropdowns unless you need JavaScript *selectize*.
- Use dropdowns or text inputs, not sliders.
- Include a data table with every chart.
- Write alt text when you write the chart code.
- Run keyboard-only and screen reader navigation tests before every release.

Thank You!

Download the accessible and non-accessible app files to practice on you own.
It's an iterative process. Small fixes compound into a dramatically better experience.

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